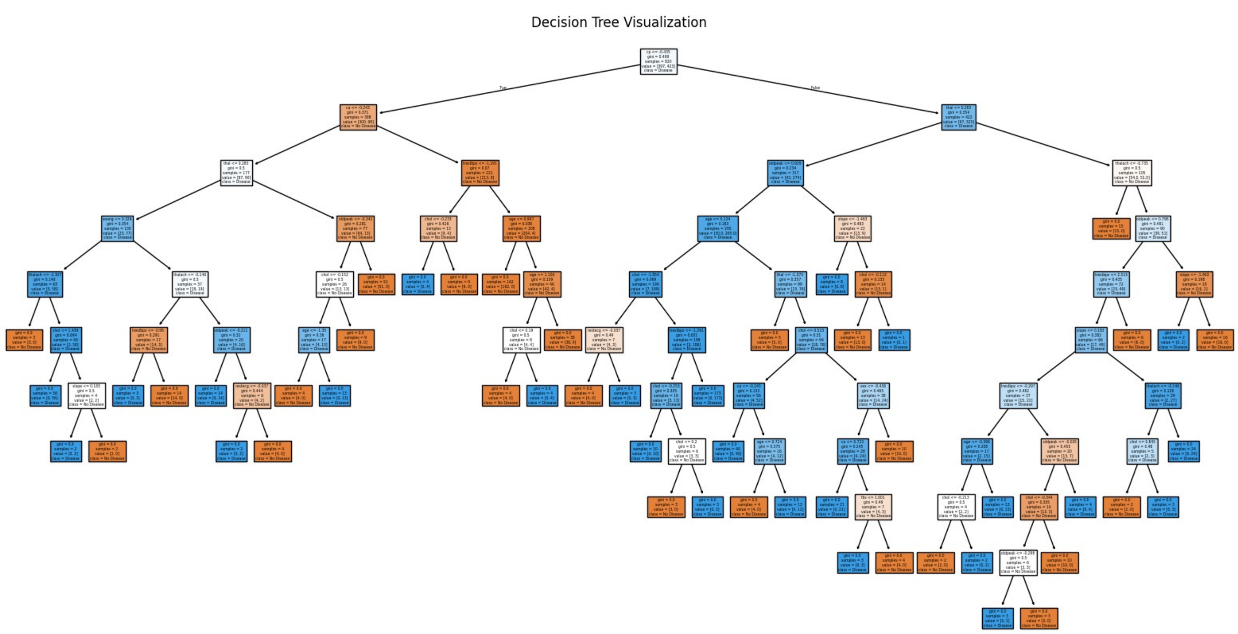
```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score, classification_report,
confusion_matrix

from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.neighbors import KNeighborsClassifier
from sklearn.neural_network import MLPClassifier


df = pd.read_csv("/content/heart.csv")
print(df.head())


print(df.isnull().sum())

X = df.drop('target', axis=1)
y = df['target']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)


scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)




dt_model = DecisionTreeClassifier(random_state=42)
dt_model.fit(X_train, y_train)
dt_pred = dt_model.predict(X_test)


plt.figure(figsize=(20, 10))
plot_tree(dt_model, feature_names=X.columns,
          class_names=['No Disease', 'Disease'], filled=True)
plt.title("Decision Tree Visualization")
plt.show()


knn_model = KNeighborsClassifier(n_neighbors=5)
knn_model.fit(X_train, y_train)
```

```
knn_pred = knn_model.predict(X_test)


mlp_model = MLPClassifier(hidden_layer_sizes=(100, 50),
                          max_iter=500, random_state=42)
mlp_model.fit(X_train, y_train)
mlp_pred = mlp_model.predict(X_test)


def evaluate_model(name, y_test, y_pred):
    print(f"----- {name} -----")
    print("Accuracy:", accuracy_score(y_test, y_pred))
    print("\nClassification Report:\n", classification_report(y_test,
y_pred))

    cm = confusion_matrix(y_test, y_pred)
    sns.heatmap(cm, annot=True, fmt='d', cmap='coolwarm')
    plt.title(name)
    plt.xlabel('Predicted')
    plt.ylabel('Actual')
    plt.show()

evaluate_model("Decision Tree", y_test, dt_pred)
evaluate_model("KNN", y_test, knn_pred)
evaluate_model("MLP", y_test, mlp_pred)

results = {
    "Decision Tree": accuracy_score(y_test, dt_pred),
    "KNN": accuracy_score(y_test, knn_pred),
    "MLP": accuracy_score(y_test, mlp_pred)
}

results_df = pd.DataFrame(list(results.items()), columns=['Model',
'Accuracy'])
print(results_df)

sns.barplot(x='Model', y='Accuracy', data=results_df)
plt.title("Model Comparison")
plt.show()
```

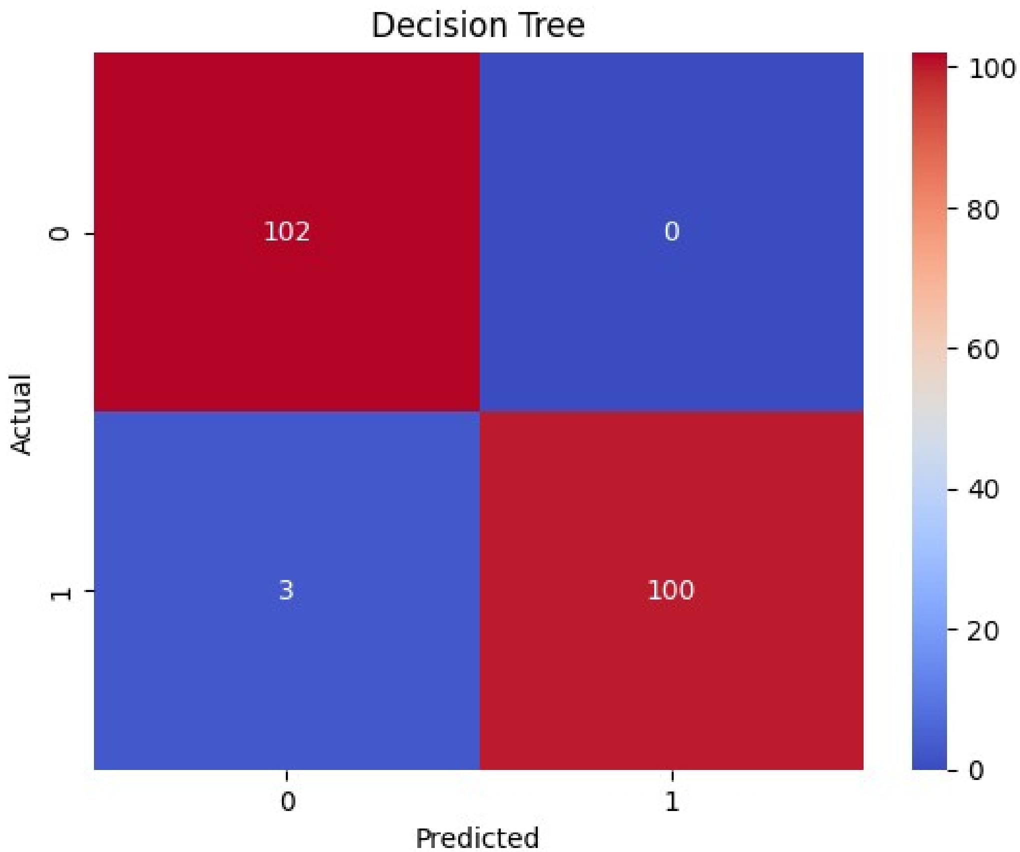
```
   age  sex  cp  trestbps  chol  fbs  restecg  thalach  exang  oldpeak  
slope  \
0   52    1   0       125   212    0        1      168      0      1.0   
2   
1   53    1   0       140   203    1        0      155      1      3.1   
0   
2   70    1   0       145   174    0        1      125      1      2.6   
0   
3   61    1   0       148   203    0        1      161      0      0.0   
2   
```

```
4   62    0   0        138   294    1         1       106       0       1.9
1

   ca  thal  target
0   2     3       0
1   0     3       0
2   0     3       0
3   1     3       0
4   3     2       0
age         0
sex         0
cp          0
trestbps    0
chol        0
fbs         0
restecg     0
thalach     0
exang       0
oldpeak     0
slope       0
ca          0
thal        0
target      0
dtype: int64
```

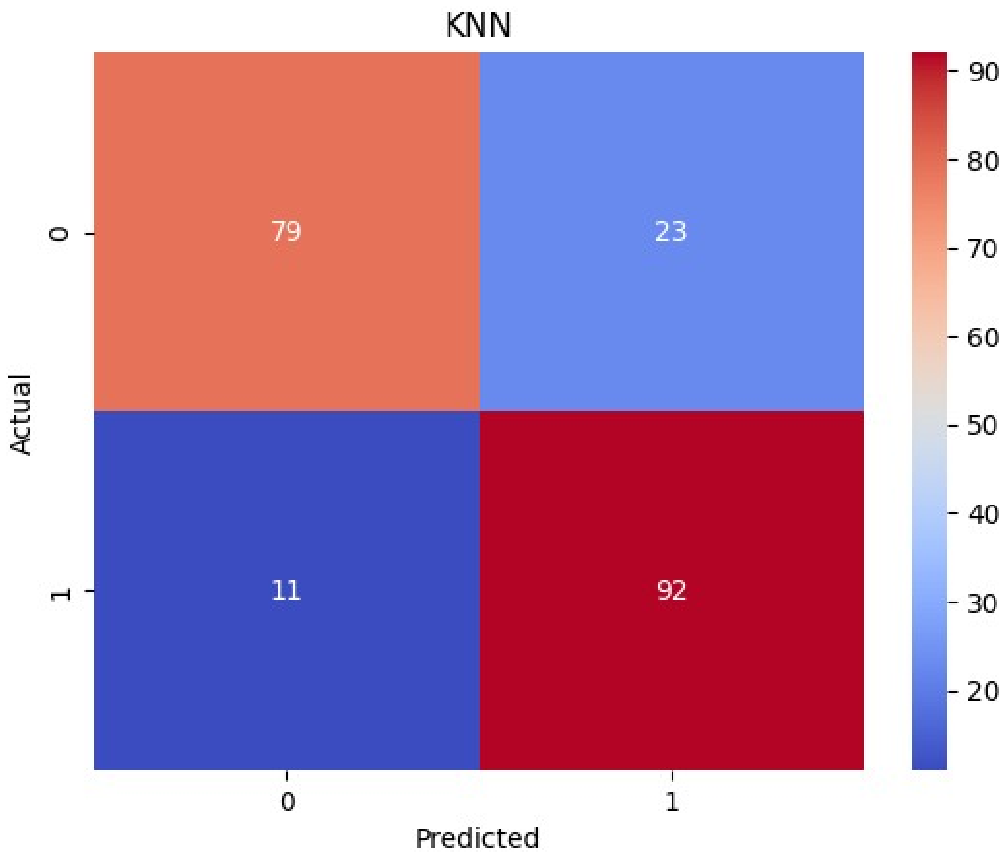



```
----- Decision Tree -----
Accuracy: 0.9853658536585366

Classification Report:
```

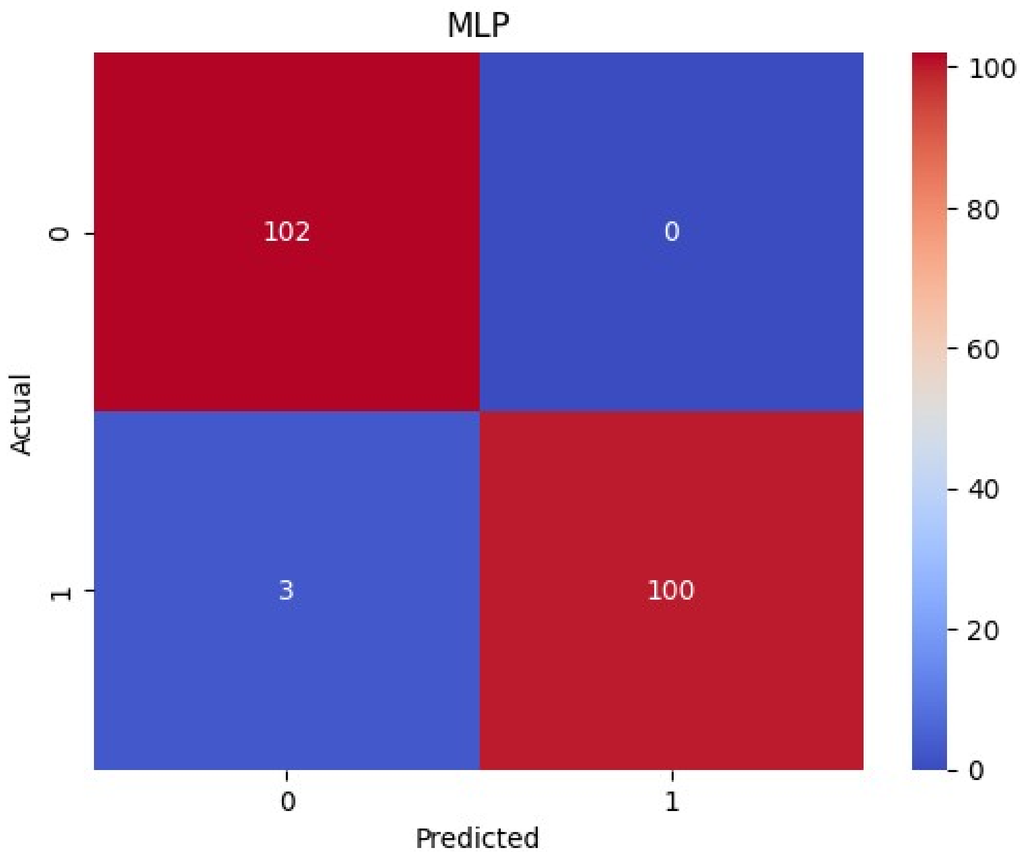
```
              precision    recall  f1-score   support

           0       0.97      1.00      0.99       102
           1       1.00      0.97      0.99       103

    accuracy                           0.99       205
   macro avg       0.99      0.99      0.99       205
weighted avg       0.99      0.99      0.99       205
```



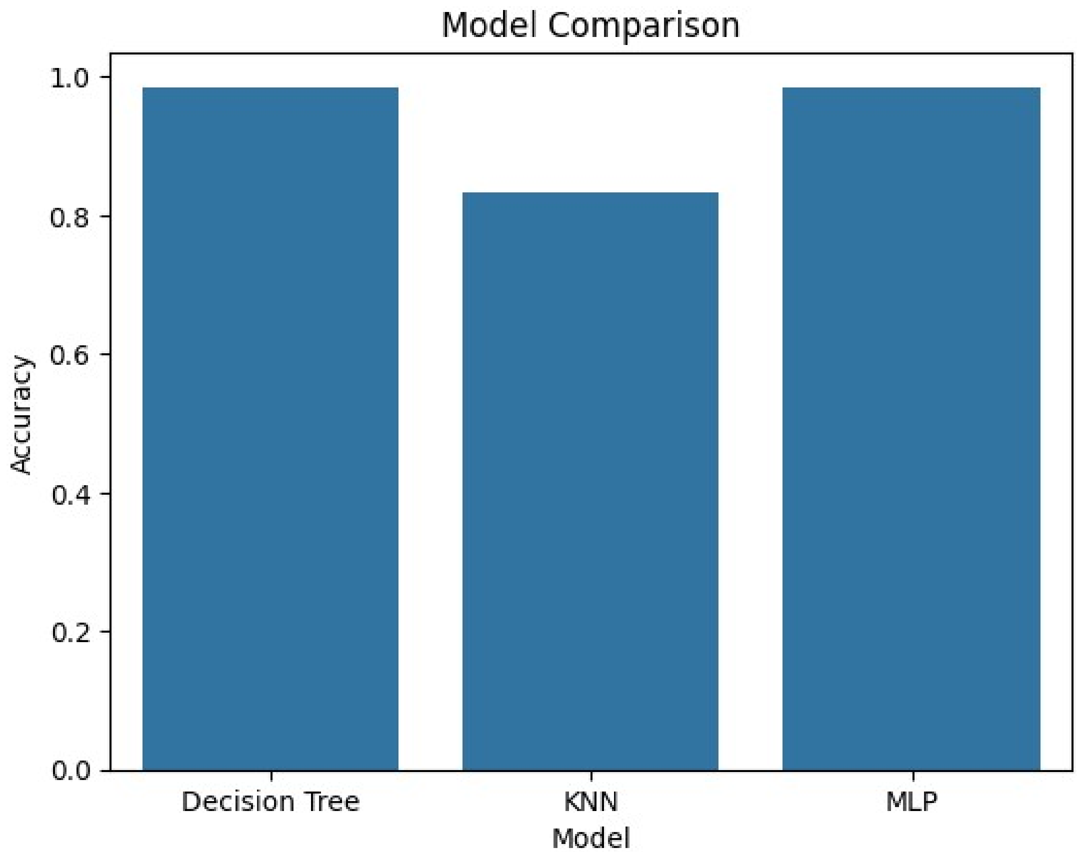

```
----- KNN -----
Accuracy: 0.8341463414634146

Classification Report:
              precision    recall  f1-score   support

           0       0.88      0.77      0.82       102
           1       0.80      0.89      0.84       103

    accuracy                           0.83       205
   macro avg       0.84      0.83      0.83       205
weighted avg       0.84      0.83      0.83       205
```




```
----- MLP -----
Accuracy: 0.9853658536585366

Classification Report:
              precision    recall  f1-score   support

           0       0.97      1.00      0.99       102
           1       1.00      0.97      0.99       103

    accuracy                           0.99       205
   macro avg       0.99      0.99      0.99       205
weighted avg       0.99      0.99      0.99       205
```



	Model	Accuracy
0	Decision Tree	0.985366
1	KNN	0.834146
2	MLP	0.985366


Model Comparison
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
Decision Tree
KNN
MLP
Model